

Preprocessing / Feature Engineering

- One-Hot Encoding
 - Label-Encoding
 - ANOVA
 - ANOVA F-test
 - Hypothesis Testing
 - Qualitative Data
 - Quantitative Data
 - Polynomial features
 - Overfitting
 - Underfitting
 - Random state
 - FEA-plots
 - Scalability - 2 types
- null-values
 outliers
 Co-variation
 chi square
 collinearity / Multi / U IF
 finding Duplicates
 PCA-Dimension Reduction
 Transformation
 Creating new columns

train-test split 80 20 (Validation)

fit transform

MSE, MAE, RMSE, R^2

$$\frac{1}{n} \sum (y - \hat{y})^2 \quad \frac{1}{n} \sum |y - \hat{y}| \quad \sqrt{\text{MSE}}$$

Confusion Matrix

Depends on Q.
Regression & Classification

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Distance

KNN

SYM

Tree

Decision tree

Random forest

Formula of
Gini & Entropy

Probability

Naive

Bayes

QDA

Boosting

Gradient Boost

AdaBoost

XGBoost

CartBoost

LightGBM

Stacking

Classifier

$$P(K|C) = \frac{P(C|X)P(X)}{P(C)}$$

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1. Tokens, n-grams
2. Stop Words
3. Stemming
4. Lemmatize
5. Count Vectorization, TFIDF Vectorisation
6. Wordnet
7. Document Term Matrix
8. Multinomial, Naive Bayes
9. Sentiment Analysis - VADER, Text Blob

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